

Dealing with Diabetes

Abstract:

Imagine if you had to take medicine every time you ate, and you had to use a math formula to figure out how much medicine to take based on the nutritional makeup of the food, how you were feeling, and what activities you were planning to do. You might also need extra medicine throughout the day (even in the middle of the night) based on your blood glucose levels. People with diabetes who take insulin do this every day. They use finger sticks and blood glucose meters or a continuous glucose sensor to monitor their blood glucose throughout the day and take insulin to keep blood glucose within safe levels. Improvements in insulin pump and glucose sensor technology have led to smarter "hybrid loop" systems that can automatically adjust and administer insulin in response to changes in blood glucose levels. Because this type of system attempts to mimic the way the human pancreas works to regulate blood glucose, it is sometimes referred to as an artificial pancreas.

In this project we will explore how these systems work by building a circuit that acts as a model of an artificial pancreas using two different liquids (tap water and distilled water instead of insulin and blood). We will try to fine-tune our circuit so it can quickly and safely adjust "blood sugar" to the right levels.

Materials required:

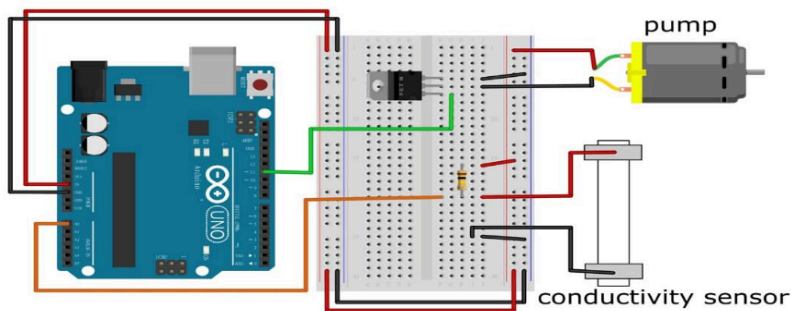
- Arduino-compatible microcontroller board
- USB cable
- Breadboard
- Jumper wires
- 5V peristaltic liquid pump
- N-channel MOSFET
- 100 k Ω resistor
- Alligator clip leads
- •Multimeter
- Bowls or food storage containers (2) 1. Tap water 2. Ionized water
- Aluminum foil
- Tape
- Corks or packing foam

Working Mechanism

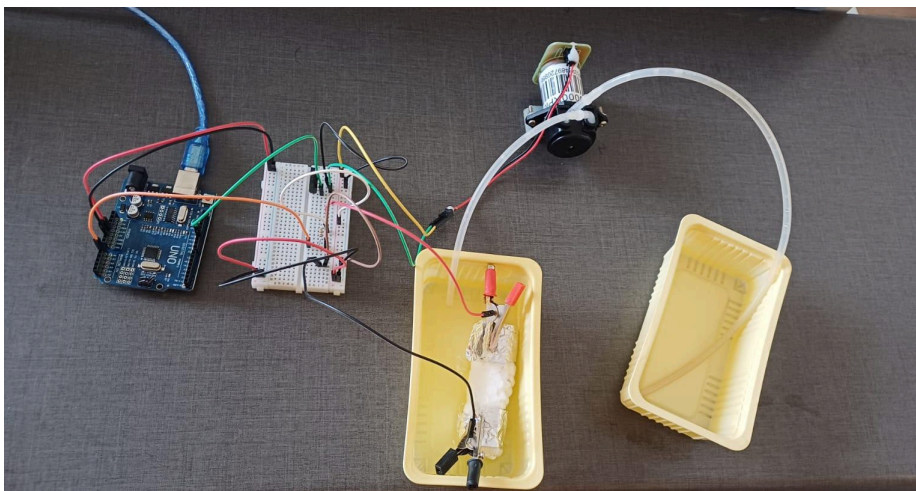
In this project, we will experiment to see how an artificial pancreas can automate insulin delivery in response to blood glucose levels by building a simplified model of an artificial pancreas. We will use two different liquids to represent blood and insulin: distilled water and tap water. Instead of measuring glucose content of the liquids, we will measure their electrical conductivity by building a circuit that outputs a voltage.

Distilled water is not very conductive. In the model, it will represent blood with high sugar content, which will cause the circuit to output a high voltage. Tap water is much more conductive, so when we add it to distilled water, the conductivity increases. This means you can use tap water to represent insulin in your model. When we add it to distilled water, the conductivity will increase, and the circuit's output voltage will drop, corresponding to a decrease in blood sugar in the model.

Circuit Diagram:



Experimental Setup:



The experimental setup, with the pump arranged to pump tap water into the container of distilled water

Working:



701	398
698	389
694	395
694	417
695	395
698	408
702	396
703	411
700	394

The serial monitor initially showed a value of 700, indicating high "blood sugar" (low conductivity). As the pump added tap water (insulin), the value dropped to 400, simulating a decrease in blood glucose. This demonstrates how insulin lowers glucose in an artificial pancreas system.

Applications:

An artificial pancreas, or insulin pump that coordinates with a continuous glucose monitor, can automate some of the maintenance and monitoring of blood glucose, including increasing or decreasing insulin, to help people control their blood glucose more effectively. These systems can take over some of the decisions and calculations involved in insulin therapy and can monitor and respond to blood glucose 24/7, often making changes every few minutes. These systems can reduce the burden of managing diabetes and may lead to better overall management.

Conclusion:

It is very important for an artificial pancreas to have a well-designed feedback control system to keep blood glucose levels in a safe range. Understanding this timing is very important to insure someone doesn't receive too much insulin and is an important consideration for scientists as they develop the algorithms used by artificial pancreas systems. The model that you will build in this project generally responds to changes very quickly, within a few seconds, but it is important to remember that real feedback control systems can take much longer depending on the physical system that is being controlled.

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